Department of Information Security and Communication Technology

TTM4115 Design of Communicating Systems

Project: Drone Airspace Advisory System (DAAS)

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Project Specification

Drone Airspace Advisory System (DAAS)

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Specification Document

1. Background

As drone-based package delivery scales, low-altitude urban airspace will become increasingly congested, with autonomous drones operating alongside helicopters, emergency services, and other manned aircraft, especially near hospitals and densely populated areas. Safe operation in such environments requires timely coordination between multiple airspace participants.

At present, there is no widely adopted mechanism for automatic, cross-operator deconfliction of drone traffic in shared low-altitude airspace. As the number of airborne actors increases, maintaining safe separation becomes more difficult, increasing the risk of conflicts, operational inefficiencies, and reduced public trust in autonomous aerial operations.

Ensuring safe and scalable package-delivery operations in such environments therefore requires reliable methods for monitoring active airspace, detecting potential conflicts, enforcing operational constraints, and coordinating drone behavior before unsafe situations occur.

2. Vision Statement

Our vision is to support safe, coordinated, and observable package-delivery drone operations in shared low-altitude urban airspace. The system should help drone operators and supervisory actors maintain situational awareness, detect unsafe situations early, and manage conflicts and airspace constraints before they lead to safety-critical events.

The prototype should provide a clear and demonstrable representation of how monitored drones can participate safely in shared urban airspace, while remaining structured and extensible enough to support additional drones, constraints, and advisory logic in future iterations.

3. Objectives

OBJ-1 — Provide shared situational awareness of active low-altitude airspace
The system shall maintain a consistent and up-to-date view of active monitored drones in shared low-altitude airspace, enabling supervisory actors to observe the current state of drone traffic.

This objective is achieved when:

- each active drone periodically publishes telemetry containing position, altitude, velocity, and flight state;
- the ATC Core updates the global airspace state when new telemetry is received; and
- the monitoring interface displays the current position and state of active drones.

Verification: The objective will be verified by running a simulation with multiple drones and confirming that telemetry updates are correctly reflected in the airspace view.

OBJ-2 — Detect potential airspace conflicts before loss of separation
The system shall identify potential loss-of-separation situations between airborne entities early enough to support conflict resolution.

This objective is achieved when:

- the ATC Core predicts short-term trajectories of relevant airborne entities;
- the system identifies situations that may lead to unsafe proximity; and
- the system produces a conflict status that can trigger advisory handling.

Verification: The objective will be verified by configuring intentionally intersecting drone trajectories and confirming that the system detects the conflict before the unsafe situation is reached.

OBJ-3 — Support observable conflict resolution for monitored drones
The system shall support the resolution of detected airspace conflicts and make the resulting drone behavior observable in the monitored airspace state.

This objective is achieved when:

- the system generates a deconfliction advisory for an affected drone;
- the drone transitions to an evasive maneuver or failsafe state; and
- the resulting trajectory change is visible in the airspace view and restores safe separation.

Verification: The objective will be verified by triggering a conflict scenario and confirming that the affected drone executes the advisory behavior and that the updated airspace view reflects the resolution.

OBJ-4 — Enforce operational constraints in monitored urban airspace
The system shall allow supervisory actors to define and activate operational constraints such as restricted zones or priority rules.

This objective is achieved when:

- restricted zones or priority rules can be defined in the system;
- these constraints are visible in the monitoring interface; and
- the constraints influence trajectory evaluation, conflict detection, or advisory generation.

Verification: The objective will be verified by defining a restricted zone and confirming that the system either prevents entry or generates behavior that avoids the restricted area.

4. Stakeholders

Stakeholders	Roles in the system	Values and interests
Project team	Designs, implements, integrates, and demonstrates the DAAS prototype as part of the course project.	Wants a project scope that is realistic within one semester. Values a stable demo, coherent documentation, architectural clarity, and a system whose behavior can be explained and defended.
Drone Operator	Registers drones for monitored flight and relies on the system to participate safely in shared airspace.	Wants lightweight registration, reliable activation, clear drone status, and predictable advisory handling. Values stable communication, mission visibility, and safe monitored flight execution.
Air Traffic Controller	Monitors active airspace, supervises drone participation, and defines operational constraints such as restricted zones or priority rules.	Wants a clear and responsive overview of active airspace, easy definition of constraints, and trustworthy conflict handling. Values timely updates, visible safety-relevant events, and explainable system behavior.
Emergency Traffic Pilot	Operates in or near the same low-altitude urban airspace as monitored drones.	Values safe separation from drones, predictable drone behavior, and reduced risk of unexpected conflicts. Is interested in reliable priority handling and protection around sensitive routes or operations.
Course staff	Assesses the quality, coherence, and demonstrability of the project.	Wants clear alignment between requirements, use cases, diagrams, and implementation. Values visible system behavior, justified design decisions, and a manageable, well-scoped project.

5. Use Case Sea-Level Overview

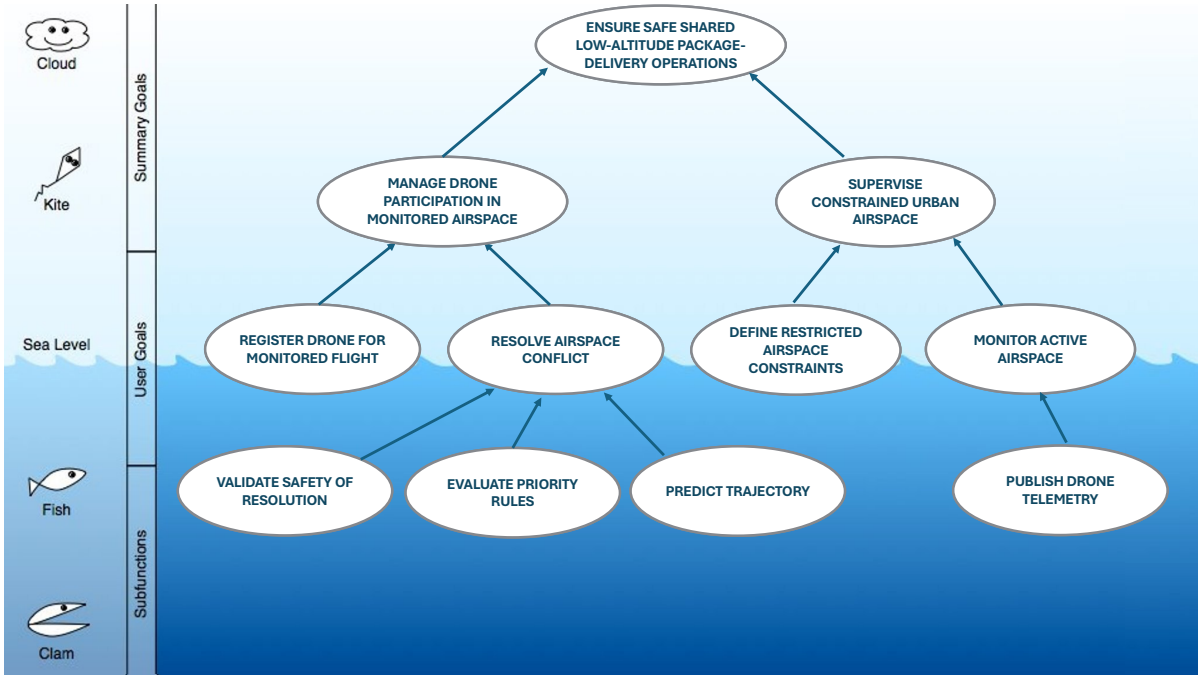

Figure 1: Use case diagram of the System.

In this document, use cases are categorized according to their level of abstraction. The prefix used in the use case identifier indicates its abstraction level:

- **UC-C** (Cloud Level): Represents high-level business goals and overall system objectives.
- **UC-K** (Kite Level): Represents strategic goals that bridge high-level objectives and user-level functionality.
- **UC-S** (Sea Level): Represents user-goal level use cases, describing complete interactions that deliver value to an actor.

This layered structure ensures traceability from high-level system objectives down to detailed functional behavior.

For the use-case tables, the system of interest is the DAAS coordination service, consisting of the ATC Core and the monitoring interface. Participating drones/drone clients interact with it as external actors.

6. Use Cases in Table Form

UC-C1: Ensure Safe Shared Low-Altitude Package-Delivery Operations

ID and Name	**UC-C1: Ensure Safe Shared Low-Altitude Package-Delivery Operations**
Created By:	Team 18 Date Created: May 4, 2026
Primary Actor	Drone Operator, Air Traffic Controller, Supervisory Authority
Secondary Actors	ATC Core System Drone Clients / Drone Simulator Web Dashboard Manned Aircraft / Emergency Traffic Data Source
Description	The system supports safe and coordinated package-delivery drone operations in monitored low-altitude urban airspace by maintaining situational awareness, enforcing operational constraints, detecting conflicts, and resolving unsafe situations before loss of separation occurs.
Trigger	A monitored package-delivery drone operation is initiated in shared urban low-altitude airspace.
Preconditions	PRE-1. The DAAS environment is deployed and operational. PRE-2. Participating drones and monitoring services exchange airspace data. PRE-3. The monitored airspace may contain constraints, priority traffic, or other airborne entities.
Postconditions	POST-1. Active drone operations are visible and coordinated in the monitored airspace. POST-2. Potential conflicts or airspace constraint violations are handled through advisories or supervisory actions. POST-3. Operational behavior and safety-relevant events are observable.
Normal Flow	1. Drone operators register drones for monitored flight participation. 2. System activates drones and receives telemetry from the airspace. 3. Air traffic controllers or supervisory authorities monitor the active airspace and define constraints when needed. 4. The system predicts trajectories and detects potential conflicts or restricted-airspace violations. 5. The system applies priority rules and generates deconfliction actions for affected drones. 6. Affected drones comply with advisories, or supervisory intervention is applied, restoring safe shared operation.
Extensions	4.E1.a Telemetry is missing or degraded → the system uses conservative estimates and marks the affected drone as degraded. 4.E1.b A restricted area or priority rule changes during operation → the system re-evaluates affected trajectories. 5.E1.a An advisory is not acknowledged or cannot be executed → the system escalates to hold, abort, or operator intervention.
Priority	Very High
Special Requirements	Safety-related behavior must be deterministic, observable, and clearly explainable during demonstrations.
Frequency of Use	Continuous during monitored package-delivery operations

UC-K1: Manage Drone Participation in Monitored Airspace

ID and Name	**UC-K1: Manage Drone Participation in Monitored Airspace**
Created By:	Team 18 Date Created: May 4, 2026
Primary Actor	Drone Operator
Secondary Actors	ATC Core System Drone Client / Drone Simulator Web Dashboard
Description	The system allows a drone operator to make drones participate in the monitored airspace by registering them, activating them for flight, and keeping their operational state visible throughout mission execution.
Trigger	Drone operator wants a drone to participate in a monitored package-delivery flight.
Preconditions	PRE-1. The DAAS environment is operational. PRE-2. MQTT communication is available between system and drone client. PRE-3. A drone client or simulator instance is available for registration and monitored flight.
Postconditions	POST-1. The drone is registered and recognized by the ATC Core. POST-2. The drone is activated for monitored flight participation. POST-3. The drone's operational state and telemetry are visible in the monitored airspace view.
Normal Flow	1. The drone operator starts a drone client or simulator instance for a monitored mission. 2. The drone client connects to the communication infrastructure and registers with the ATC Core. 3. The ATC Core validates the drone identity and stores the drone as a known airspace participant. 4. The ATC Core activates the drone for monitored flight and enables its operational channels. 5. The drone begins mission execution and periodically publishes telemetry. 6. The ATC Core updates the global airspace picture with the drone's current state. 7. The Web Dashboard presents the drone as an active participant in the monitored airspace.
Extensions	3.E1.a Registration data is invalid → registration is rejected and the operator must retry with corrected data. 3.E1.b The drone identifier is already in use → the system requests re-registration or a new identifier. 4.E1.a The drone cannot be activated or does not confirm operational mode → the system marks it as inactive and delays mission participation. 5.E1.a Telemetry is interrupted during operation → the system marks the drone as degraded or inactive in the airspace view.
Priority	High
Special Requirements	Drone participation management must be lightweight, deterministic, and support repeated registration/activation during demonstrations.
Frequency of Use	Once per drone session, then continuously during monitored flight

UC-K2: Supervise Constrained Urban Airspace

ID and Name	**UC-K2: Supervise Constrained Urban Airspace**
Created By:	Team 18 Date Created: May 4, 2026
Primary Actor	Air Traffic Controller , Supervisory Authority
Secondary Actors	ATC Core System Web Dashboard Drone Clients / Drone Simulator
Description	The system allows supervisory actors to oversee active low-altitude drone operations in constrained urban airspace by monitoring traffic, defining operational constraints, and ensuring that airspace rules influence conflict handling and flight behavior.
Trigger	A supervisory actor needs to oversee ongoing drone operations or apply operational constraints in monitored urban airspace.
Preconditions	PRE-1. The DAAS environment is operational. PRE-2. Active or pending drone operations exist in the monitored airspace. PRE-3. Supervisory actors can access the airspace monitoring interface.
Postconditions	POST-1. The current airspace state is visible to supervisory actors. POST-2. Operational constraints such as restricted zones or priority rules are defined and active in the system. POST-3. Active constraints influence conflict detection and resolution behavior.
Normal Flow	1. A supervisory actor opens the monitoring interface for the active airspace. 2. The system displays active drones, their current positions, and relevant flight states. 3. The supervisory actor reviews the current operational situation and identifies the need for a constraint or rule. 4. The supervisory actor defines a restricted airspace area or priority rule. 5. The ATC Core stores and activates the constraint in the monitored airspace model. 6. The updated constraint becomes visible in the airspace view. 7. The ATC Core applies the active constraint during trajectory evaluation, conflict detection, and advisory generation.
Extensions	4.E1.a The defined constraint is incomplete or invalid → the system rejects it and requests correction. 5.E1.a The constraint cannot be activated consistently → the system keeps the previous valid airspace configuration and alerts the supervisory actor. 7.E1.a A drone is predicted to violate an active constraint → the system generates a warning or advisory to prevent entry into the restricted area.
Priority	High
Special Requirements	Constraint definitions must be visible, auditable, and immediately reflected in the monitored airspace representation used for supervision and conflict handling.
Frequency of Use	Occasional during preparation and supervision of active urban airspace operations

UC-S1: Register Drone for Monitored Flight

ID and Name	**UC-S1: Register Drone for Monitored Flight**
Created By:	Team 18 Date Created: May 4, 2026
Primary Actor	Drone Operator
Secondary Actors	Drone Client / Drone Simulator
Description	A drone operator registers a drone for participation in a monitored mission so that the drone becomes an active and visible participant in the monitored airspace.
Trigger	The drone operator starts a drone client or simulator instance for a monitored mission.
Preconditions	PRE-1. The DAAS environment is operational. PRE-2. MQTT communication is available between the drone client and the system. PRE-3. A valid drone identity and mission context are available.
Postconditions	POST-1. The drone is registered as a known participant in monitored airspace. POST-2. The drone is activated for monitored flight participation. POST-3. The drone appears as an active entity in the airspace view.
Normal Flow	1. The drone operator starts the drone client or simulator. 2. The drone client connects to the MQTT broker. 3. The drone client publishes a registration request. 4. The system validates the drone identity and registration data. 5. The system stores the drone as a known airspace participant. 6. The system activates the drone for monitored flight and enables telemetry/advisory channels. 7. The drone enters its initial operational state and becomes visible in the airspace view.
Extensions	4.E1.a Registration data is invalid → registration is rejected and the operator must retry with corrected data. 4.E1.b The drone ID is already in use → the system requests re-registration or a new identifier. 6.E1.a The drone does not confirm operational mode → the system marks the drone as inactive and delays mission participation.
Priority	High
Special Requirements	Registration and activation must be lightweight, deterministic, and auditable during demonstrations.
Frequency of Use	Once per drone session

UC-S2: Resolve Airspace Conflict

ID and Name	**UC-S2: Resolve Airspace Conflict**
Created By:	Team 18 Date Created: May 4, 2026
Primary Actor	Air Traffic Controller / Supervisory Authority
Secondary Actors	Participating Drone / Drone Client Other Airborne Entity
Description	A supervisory actor resolves a predicted airspace conflict between monitored airborne entities so that safe separation is restored before unsafe proximity occurs.
Trigger	The system identifies a potential loss-of-separation situation in monitored airspace.
Preconditions	PRE-1. At least two relevant airborne entities are active in monitored airspace. PRE-2. Airspace state data is available with sufficient quality for conflict assessment. PRE-3. The affected drone can receive and execute advisories or failsafe actions.
Postconditions	POST-1. A conflict resolution action has been selected and issued. POST-2. The affected drone has executed an avoidance maneuver or a failsafe alternative. POST-3. Safe separation is restored or the situation is escalated for further supervisory handling.
Normal Flow	1. The system detects a potential airspace conflict and presents the situation to the supervisory actor. 2. The system predicts short-term trajectories and evaluates the severity of the conflict. 3. The system applies priority rules to determine which entity must adapt. 4. The system selects an appropriate deconfliction strategy. 5. The system issues a collision-avoidance advisory to the affected drone. 6. Affected drone acknowledges the advisory and transitions to an evasive state. 7. The affected drone executes the required maneuver. 8. The system validates that safe separation has been restored. 9. The supervisory actor observes the updated airspace state and the system resumes normal operation.
Extensions	5.E1.a The advisory is not acknowledged → the system escalates to hold, abort, or supervisory intervention. 6.E1.a The drone cannot execute the maneuver → the drone performs a failsafe action and the system marks the situation as degraded. 8.E1.a Safe separation is not restored after the first maneuver → the system re-evaluates the conflict and issues a new resolution action.
Priority	Very High
Special Requirements	Conflict resolution must be deterministic, safety-oriented, and clearly visible in the monitoring interface during demonstrations.
Frequency of Use	Occasional, safety-critical, during active flight operations

UC-S3: Define Restricted Airspace Constraints

ID and Name	**UC-S3: Define Restricted Airspace Constraints**
Created By:	Team 18 Date Created: May 4, 2026
Primary Actor	Air Traffic Controller / Supervisory Authority
Secondary Actors	Web Dashboard
Description	A supervisory actor defines restricted airspace constraints so that monitored drone operations respect temporary no-fly zones or controlled urban airspace rules.
Trigger	A supervisory actor needs to introduce or update an operational airspace constraint.
Preconditions	PRE-1. The DAAS environment is operational. PRE-2. The supervisory actor has access to the airspace management interface. PRE-3. A monitored airspace model is available in the system.
Postconditions	POST-1. A new or updated airspace constraint is stored in the system. POST-2. The constraint is visible in the airspace view. POST-3. The constraint influences subsequent trajectory evaluation, conflict detection, and advisory generation.
Normal Flow	1. The supervisory actor opens the airspace constraint management view. 2. The system displays the current monitored airspace and existing constraints. 3. The supervisory actor defines a restricted area or updates an existing one. 4. The system validates the submitted constraint definition. 5. The system stores and activates the constraint in the airspace model. 6. The updated constraint becomes visible in the dashboard. 7. The system applies the active constraint to future airspace evaluations.
Extensions	4.E1.a The constraint definition is incomplete or invalid → the system rejects it and requests correction. 5.E1.a The constraint conflicts with the current airspace configuration → the system keeps the previous valid configuration and reports the issue. 7.E1.a An active drone is predicted to violate the new constraint → the system raises a warning or generates an advisory for avoidance.
Priority	High
Special Requirements	Constraint definitions must be immediately visible, persistent during operation, and auditable after the demonstration.
Frequency of Use	Occasional during setup and supervision of monitored operations

UC-S4: Monitor Active Airspace

ID and Name	**UC-S4: Monitor Active Airspace**
Created By:	Team 18 Date Created: May 4, 2026
Primary Actor	Air Traffic Controller / Supervisory Authority
Secondary Actors	Drone Operators
Description	A supervisory actor monitors active drone operations in real time so that the current airspace situation, relevant constraints, and safety-relevant changes remain visible and understandable.
Trigger	The supervisory actor opens or refreshes the active airspace monitoring view.
Preconditions	PRE-1. The DAAS environment is operational. PRE-2. At least one drone is registered or active in monitored airspace. PRE-3. Telemetry data can be received from active drones.
Postconditions	POST-1. The supervisory actor has an up-to-date view of active drones and their states. POST-2. Relevant airspace constraints and safety-relevant events are visible in context. POST-3. The monitoring view reflects the latest accepted telemetry and conflict status.
Normal Flow	1. The supervisory actor opens the active airspace dashboard. 2. The system retrieves the current airspace state. 3. Active drones are displayed with position, altitude, and flight state. 4. The system continuously receives telemetry updates from active drones. 5. The displayed airspace view is refreshed to reflect the latest accepted data. 6. The supervisory actor observes airspace developments, including active constraints and detected conflicts.
Extensions	4.E1.a Telemetry from a drone is missing or delayed → the system marks the drone as degraded or inactive in the dashboard. 4.E1.b Telemetry arrives out of order → the system discards stale updates based on timestamp or sequence number. 6.E1.a A conflict or restricted-airspace issue is detected while monitoring → the system highlights the affected entities and presents the safety-relevant status.
Priority	High
Special Requirements	The monitoring view must remain readable, responsive, and consistent under continuous telemetry updates from multiple drones.
Frequency of Use	Continuous during active monitored operations

7. Deployment Diagram

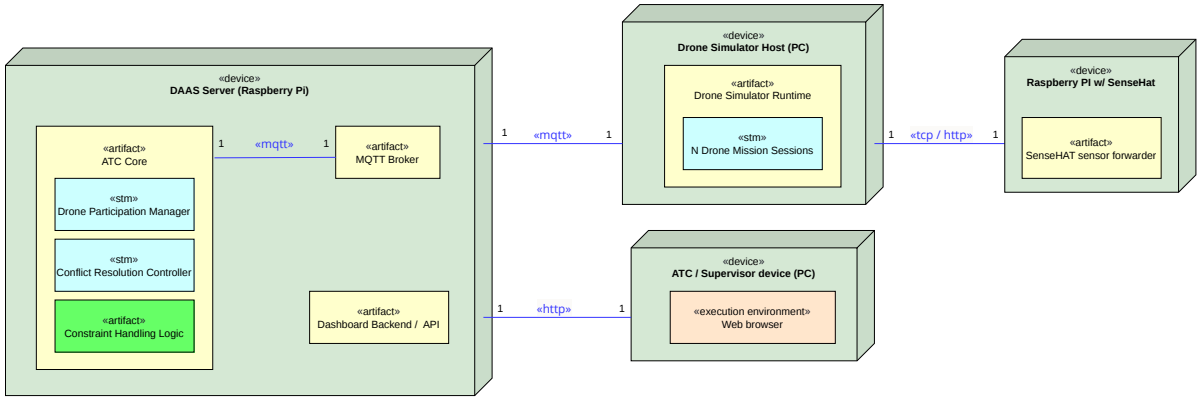

Figure 2: Deployment Diagram of the System.

Devices and Artifacts

The deployment diagram shows the physical distribution of the DAAS prototype and the communication paths between its main devices and software artifacts.

- **DAAS Server (Raspberry Pi):** This device is the central coordination node of the prototype. It hosts the *ATC Core*, the *MQTT Broker*, and the *Dashboard Backend / API*. The server receives telemetry from drones, maintains the shared airspace state, evaluates conflicts and constraints, and provides the data required by the monitoring interface.
- **Drone Simulator Host:** This device runs the *Drone Simulator Runtime*, which manages one or more simulated drone mission sessions. These drones communicate with the DAAS Server through MQTT, which is used for telemetry publication, advisory delivery, acknowledgements, and runtime coordination.
- **ATC / Supervisor Device:** This device provides access to the monitoring interface through a web browser. The browser communicates with the *Dashboard Backend / API* over HTTP and allows the supervisory actor to observe active drones, conflict states, and restricted-airspace constraints.
- **Raspberry Pi with Sense HAT:** This device runs the *SenseHAT Sensor Forwarder* and is connected to the Drone Simulator Host over TCP or HTTP. It acts as an external input source for the simulator and allows physical input to influence the simulated drone behavior.
- **State-machine-based components:** The deployment also reflects the main behavioral components implemented with state machines. The *ATC Core* contains the *Drone Participation Manager* and the *Conflict Resolution Controller*, while the *Drone Simulator Runtime* contains one or more *Drone Mission Sessions*. These state-machine-based components support reactive behavior and event-driven coordination in the system.
- **Architectural rationale:** The deployment structure separates coordination, simulation, supervision, and external input across different nodes. This keeps the system easy to demonstrate while still representing a distributed communicating system with multiple devices, protocols, and interacting software artifacts.

8. Sequence Diagram

The following sequence diagrams were developed for the system's sea-level use cases, as these represent the main user goals and therefore provide the most relevant interaction scenarios for explaining how the distributed components of the prototype communicate and coordinate.

SD-1: Register drone for monitored flight

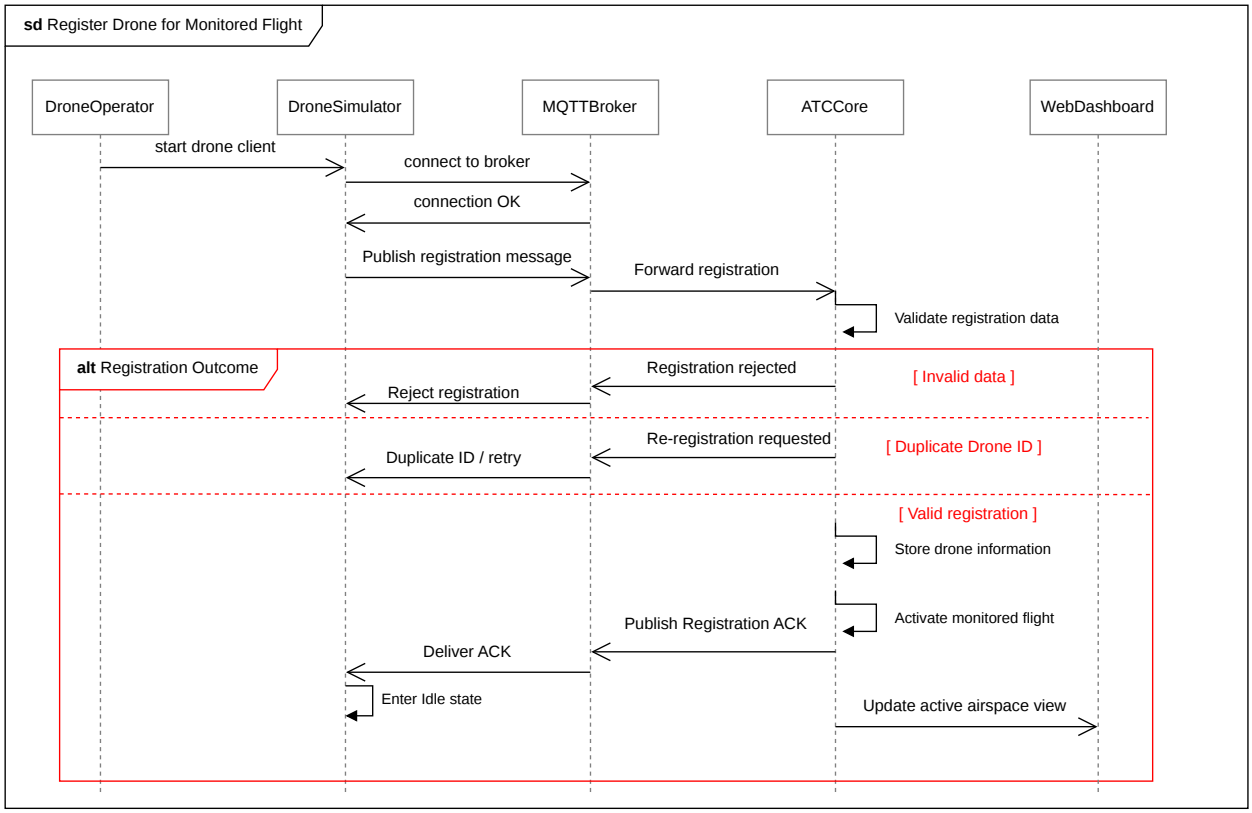

Figure 3: Sequence diagram UC-S1

SD-2: Resolve airpsace conflict

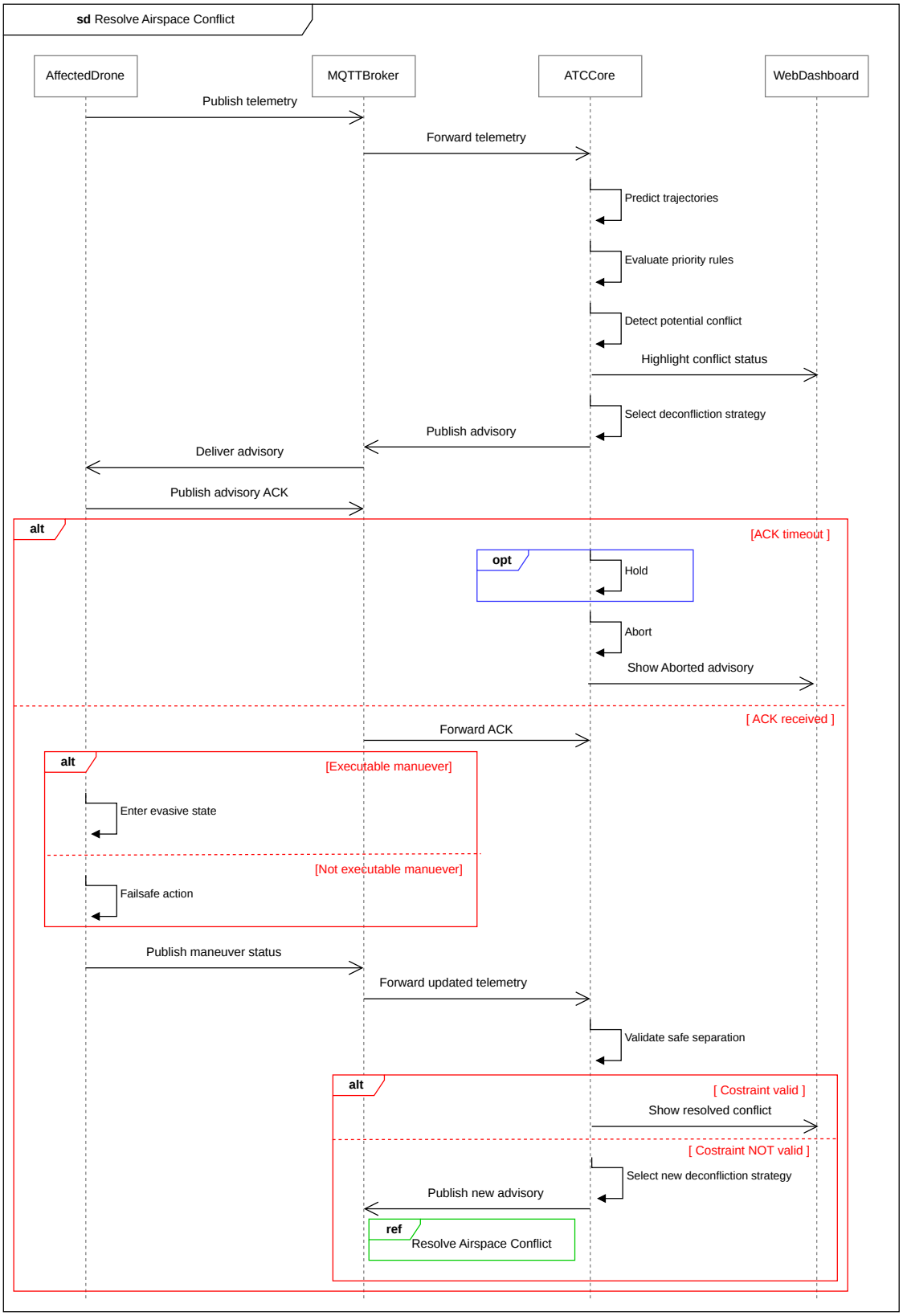

Figure 4: Sequence diagram UC-S2

SD-3: Define restricted airspace constraints

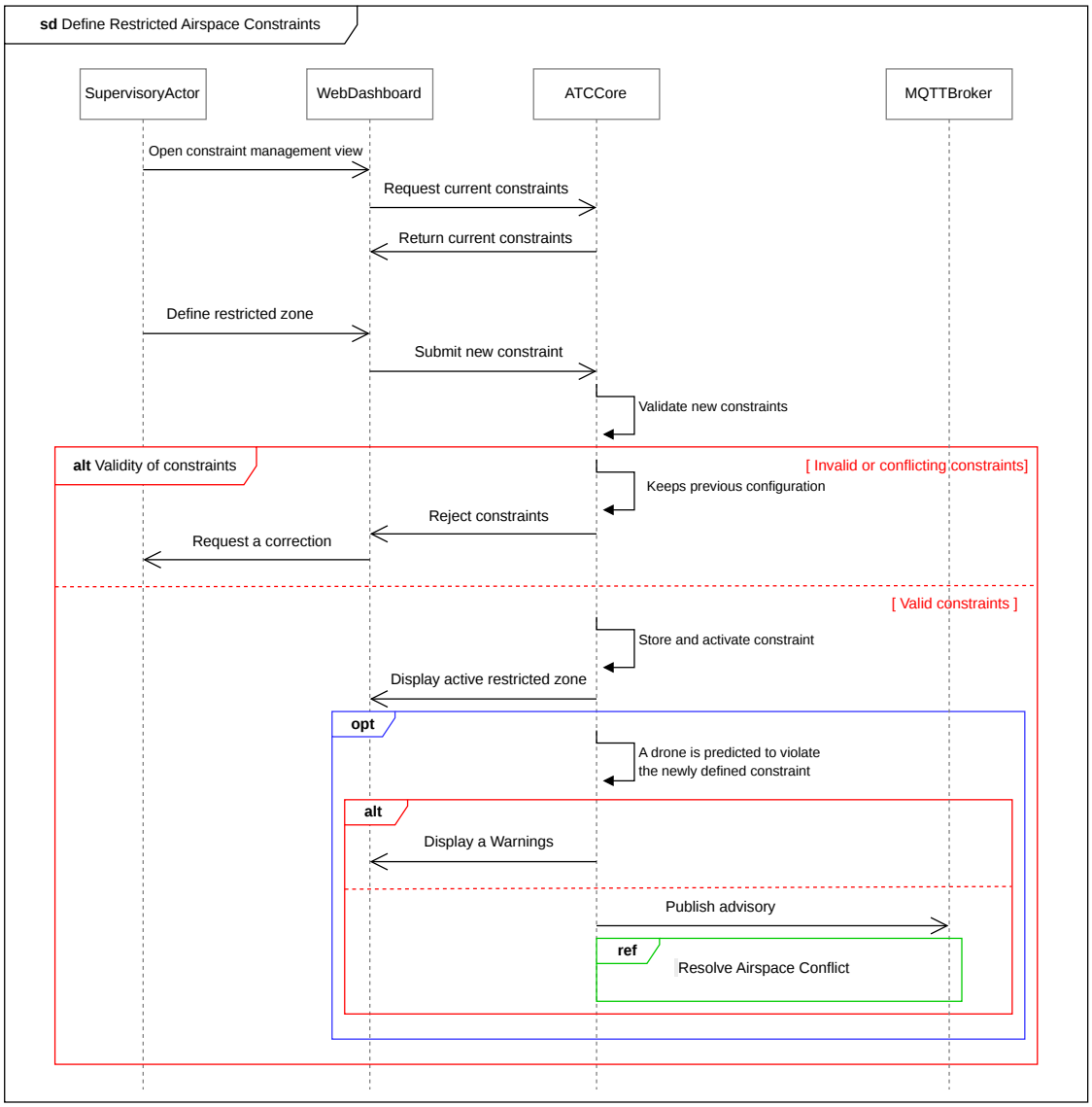

Figure 5: Sequence diagram UC-S3

SD-4: Monitored active airspace

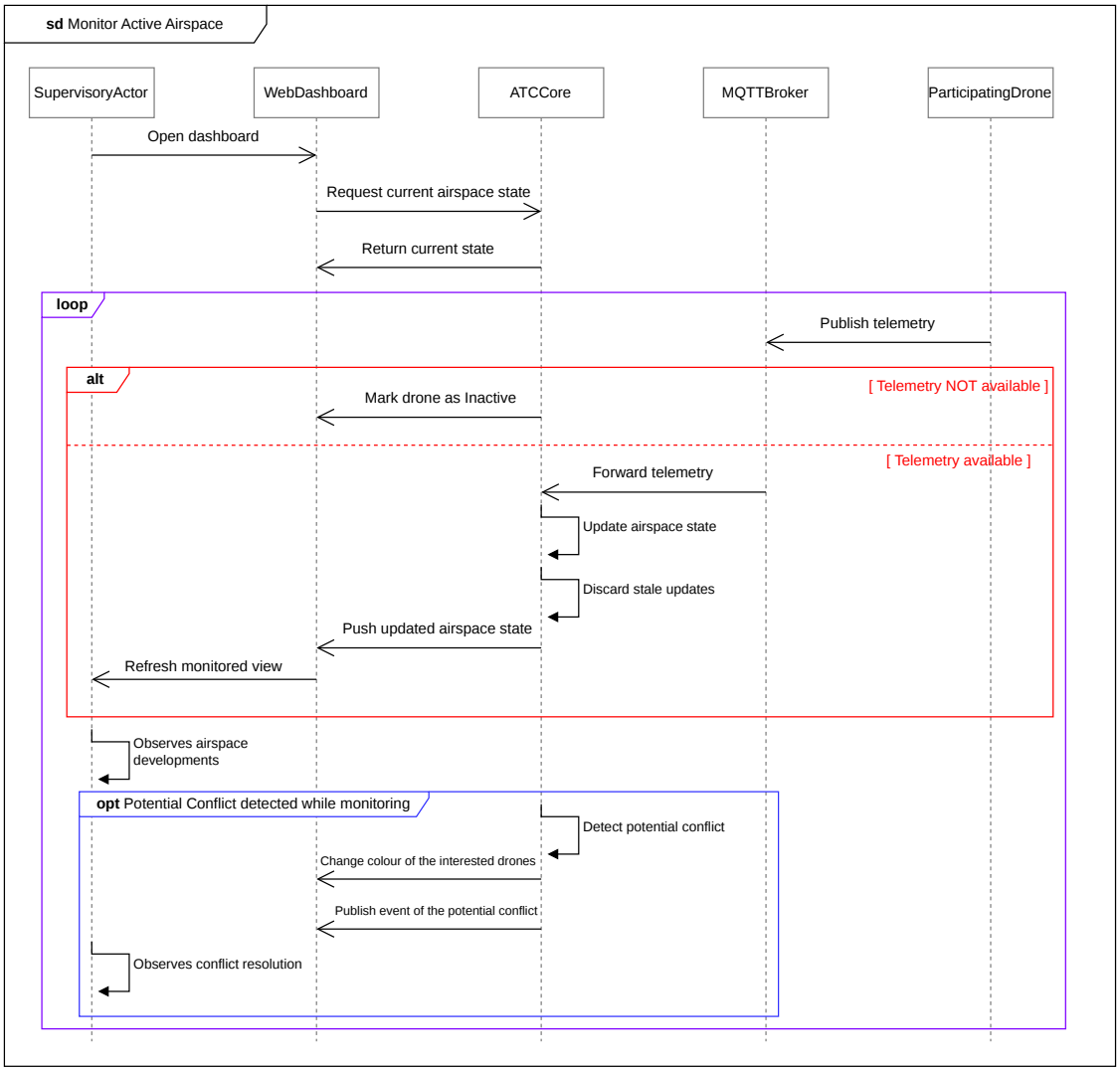

Figure 6: Sequence diagram UC-S4

9. State Machine Diagrams

The following state machines model the main reactive components of the DAAS prototype and were selected because they represent the most relevant event-driven behavior in the system. They are aligned with the selected sea-level use cases and sequence diagrams, and they provide an implementable behavioral view of how the system handles drone participation, conflict resolution, and drone mission progression. The models were designed so that they can be translated into state-machine-based logic in STMPY.

STM-1: Drone Participation Manager

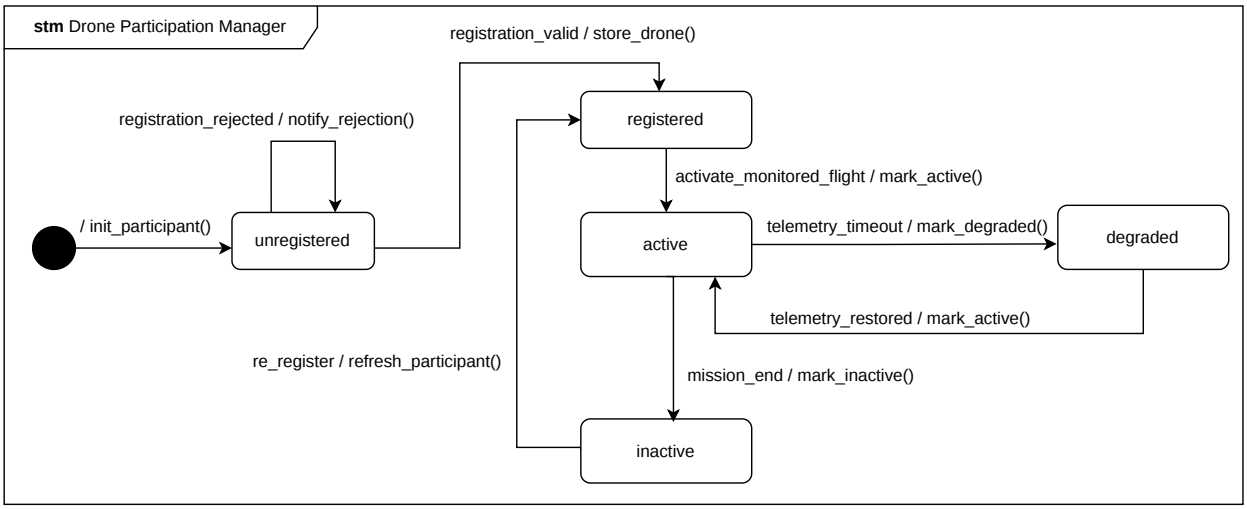

Figure 7: stm Drone Participation Manager

STM-2: Conflict Resolution Controller

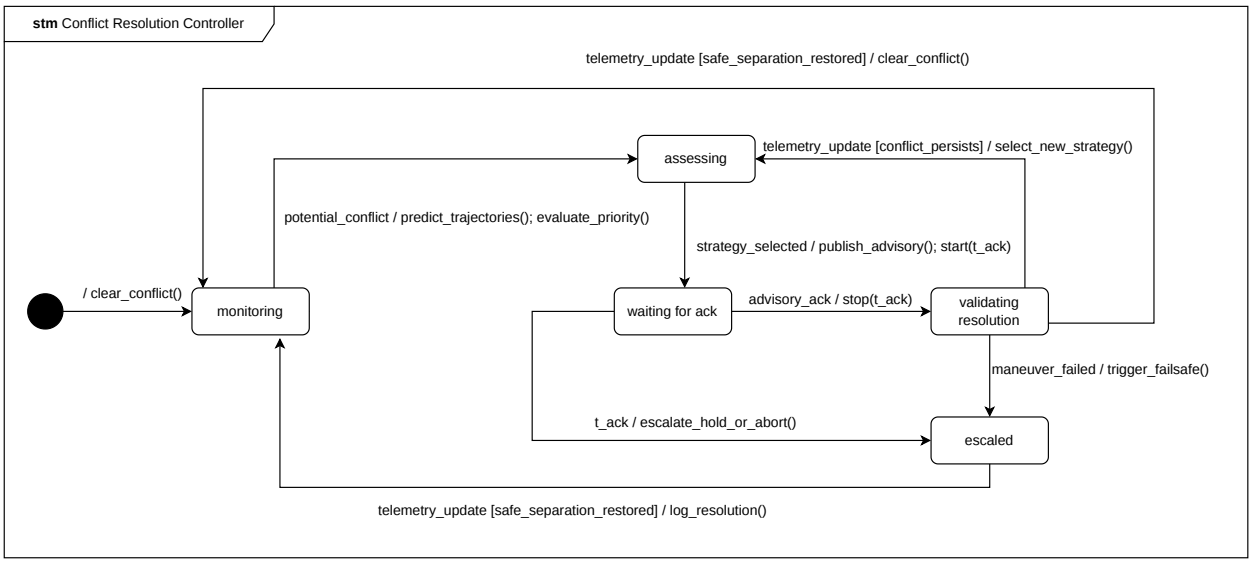

Figure 8: stm Conflict Resolution Controller

STM-3: Drone Mission Session

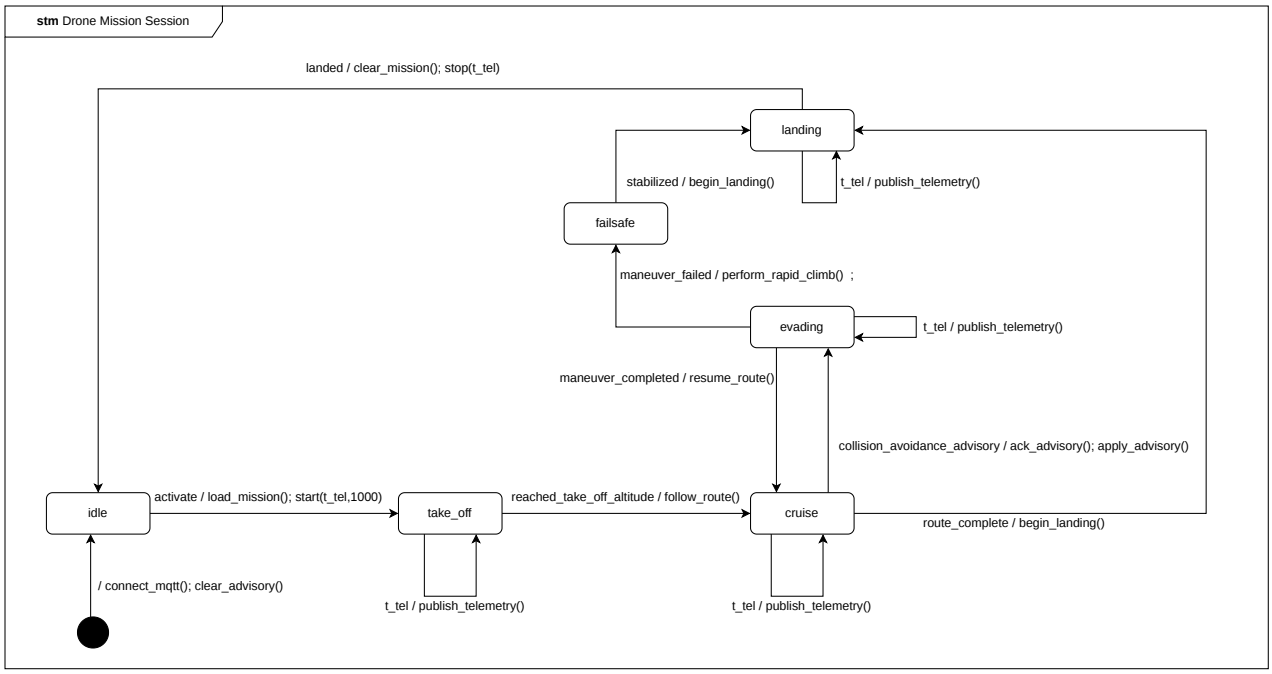

Figure 9: stm Drone Mission Session

10. Implementation Comments

Most of the demonstrated DAAS prototype works end-to-end as a distributed system. In the current implementation, drones can register through MQTT, receive monitored-flight activations, publish telemetry, and appear in the live dashboard. The airspace core maintains the shared airspace state, handles mission requests, stores and updates restricted zones, and periodically evaluates drone telemetry to generate advisories. The dashboard provides a working operator interface for monitoring the airspace, creating new mission requests from pickup and dropoff points, and managing no-fly zones. In addition, the project includes a control gateway that can forward external control inputs to a manually controlled drone session, which makes the system suitable for demonstration with Raspberry Pi / Sense HAT integration.

The parts of the system we are most proud of are the overall component integration and the fact that the main demo scenarios work across multiple communicating components instead of only as isolated modules. In particular, we want to highlight the full mission-request flow from the dashboard to the airspace core and simulator, the live monitoring interface with MQTT-to-browser updates, and the advisory handling behavior in the simulator. The simulator is able to react to conflict and restricted-zone advisories, perform route changes, and resume nominal behavior when the airspace situation becomes safe again. We also consider the modular structure of the codebase a strong result, since it allowed the project to be developed in parallel across the team.

The main project objectives defined earlier in the specification have also been verified in the implemented prototype. Shared situational awareness, conflict detection, observable conflict resolution, and operational constraint handling are all supported by the running system and are demonstrated in the attached screencast. The screencast shows that the implemented prototype is able to realize the main interaction flows described in the specification and to make the resulting system behavior visible in a concrete demonstration setting.

At the same time, some parts are still simplified compared to the full behavioral intent of the specification. In particular, the conflict-resolution logic is implemented as a periodic monitoring and advisory loop rather than as a fully explicit acknowledgement/escalation workflow, and some monitoring-related edge cases such as stale telemetry handling are represented more lightly in the code than in the analysis models. For this reason, we consider the prototype functionally strong for the demonstrated scenarios, while still leaving room for future refinement in the safety logic and supervisory workflows.

Project Repository

The implementation of the DAAS prototype is available in the following GitHub repository:

`https://github.com/FFedericoFantini/daas-course-project`